

December 2024

Duration: 3 Hours

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

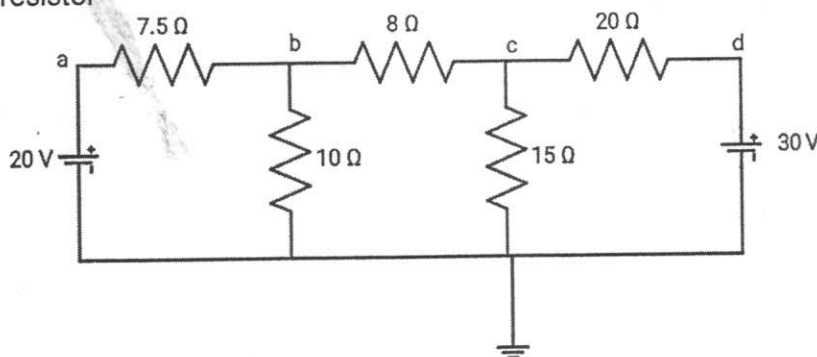
11. Which region in a BJT is the middle layer that controls the flow of charge carriers?
 A) Emitter B) Base
 C) Collector D) Gate
12. In an N-channel enhancement-mode MOSFET, what happens to the drain current (I_D) when the gate-source voltage (V_{GS}) is below the threshold voltage (V_{th})?
 A) I_D equals the gate current (I_G). B) I_D increases rapidly.
 C) I_D becomes negative. D) I_D remains zero or very small.
13. A Zener diode is specifically designed to operate in the _____.
 A) Forward bias region only B) Both forward and reverse bias regions
 C) Reverse bias region only D) Neither forward nor reverse bias regions
14. The efficiency of a full-wave bridge rectifier compared to a half-wave rectifier is _____.
 A) Lower B) The same
 C) Higher D) Cannot be compared
15. MOSFET is a _____ device
 A) Current controlled B) Voltage controlled voltage source
 C) Voltage controlled D) Voltage controlled current source
16. The r.m.s value of load current in a half wave rectifier is _____.
 A) $0.5I_m$ B) $0.5I_m$
 C) $0.707I_m$ D) $0.3183I_m$
17. A non-inverting closed loop Op-Amp circuit generally has a gain factor
 A) Less than one B) Of zero
 C) Greater than one D) Equal to one
18. The charging time or ON time of the capacitor in Astable mode of operation using IC 555 timer is
 A) $T_{ON} = 0.693(R_1 + R_2)C$ B) $T_{ON} = 0.693(R_1)C$
 C) $T_{ON} = 0.693(R_2)C$ D) $T_{ON} = 0.693(F)C$
19. Voltage gain of an Op-Amp is expressed in _____.
 A) db B) Variable
 C) Volts D) Volts/ sec
20. The main application of a 555 IC is _____.
 A) Counter B) Comparator
 C) Timer D) Amplifier

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Using nodal analysis calculate the current flowing through 10Ω resistor



- b) Discuss briefly the working principle of DC motor. Also mention the significance of back emf in DC motor
- c) Define average value of an alternating quantity. Derive an expression for average value of an alternating current.

5	L*3	1	1,2
7	L2	2	1,2
4	L2	1	1,2

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|-------|--|---|----|---|-----|
| 2. a) | A 50 kVA, 50 Hz, 1100/220 V single phase transformer the iron and full load copper losses at full load are 350W and 425W respectively. Find its efficiency at full load unity p.f. Also calculate the number of turns in primary and secondary if the maximum value of flux linking the core is 0.05 Wb. | 5 | L3 | 2 | 1,2 |
| b) | With neat circuit diagram and waveforms derive the relationship between voltage and current in a single phase RC circuit. Also derive the expression for power consumed. | 7 | L2 | 1 | 1,2 |
| c) | Draw the torque vs armature current and speed vs armature current characteristics of DC shunt motor | 4 | L1 | 2 | 1,2 |
| 3. a) | A pure inductive coil allows a current of 10A to flow from a 230V, 50 Hz supply. Find (i) inductance reactance (ii) inductance of the coil and (iii) equations for voltage and current. | 5 | L3 | 1 | 1,2 |
| b) | Prove that stator magnetic field has constant magnitude and it rotates at synchronous speed in a three-phase induction motor | 7 | L2 | 2 | 1,2 |
| c) | Define power, energy, power factor and reactive power | 4 | L1 | 1 | 1,2 |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Discuss the operation of Zener diode as a voltage regulator | 5 | L2 | 3 | 1,2 |
| b) | Discuss the construction and operation of BJT | 7 | L2 | 4 | 1,2 |
| c) | Draw the ideal, simplified and piecewise linear equivalent circuit of diode | 4 | L1 | 3 | 1,2 |
| 5. a) | A transistor has $I_B = 0.08\text{mA}$ and $I_E = 9.60\text{mA}$. Determine its collector current I_C , α , and β . | 5 | L3 | 4 | 1,2 |
| b) | With a neat circuit diagram discuss the operation of center tapped transformer full wave rectifier. Derive an expression for average DC current. | 7 | L2 | 3 | 1,2 |
| c) | Discuss briefly the operation of MOS inverter | 4 | L2 | 4 | 1,2 |
| 6. a) | A half wave rectifier has an input of 10V (peak) fed from a step-down transformer and is connected to a load of 1k Ω . Calculate DC voltage and load current for ideal and silicon diode | 5 | L3 | 3 | 1,2 |
| b) | Discuss the construction and operation of an n-channel Enhancement type MOSFET | 7 | L2 | 4 | 1,2 |
| c) | Draw the symbol and I-V characteristics of a semiconductor diode, specifying the regions of operation | 4 | L1 | 3 | 1,2 |

Unit – III

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|-------|---|---|----|---|-----|
| 7. a) | For an IC 555 timer operating as astable multivibrator has, $T_{ON} = 3\text{sec}$, $T_{OFF} = 1\text{sec}$ and $C = 10\mu\text{F}$. Calculate the duty cycle and values of R_1 and R_2 . | 5 | L3 | 5 | 1,2 |
| b) | Discuss briefly the pin description and applications of 78XX series IC voltage regulators. | 7 | L2 | 5 | 1,2 |
| c) | List the characteristics of an Op-Amp. | 4 | L1 | 5 | 1,2 |
| 8. a) | With a neat diagram discuss, operation of a summing amplifier circuit using an operational amplifier | 5 | L2 | 5 | 1,2 |
| b) | Discuss the operation of 555 Timer as astable multivibrator. | 7 | L2 | 5 | 1,2 |
| c) | An inverting amplifier has $R_1 = 20\text{k}\Omega$ and $R_f = 100\text{k}\Omega$. Find the output voltage and input current for an input voltage of 1V. | 4 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
